

Shae McFadden

PH.D. CANDIDATE · ARTIFICIAL INTELLIGENCE & CYBERSECURITY

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Summary

Currently a Ph.D. candidate, supervised by Dr. Fabio Pierazzi and Dr. Nicola Paoletti, at King's College London, investigating the applications of deep reinforcement learning to cybersecurity. A passion for the cross-section between artificial intelligence and cybersecurity resulted in collaborations with the Systems Security Research Lab (S2Lab) at UCL and the AI for Cyber Defence Research Centre (AICD) at The Alan Turing Institute. Graduated from King's College London with a first-class B.Sc. (Hons) in Computer Science specialising in Artificial Intelligence. During undergraduate studies, was awarded the Layton Research Award, the Alan Fairbourn Memorial Prize, and the Associateship of King's College.

Experience

Ph.D. Candidate in Computer Science

KING'S COLLEGE LONDON

London, UK
October 2023 - Present

- Thesis on "The Applications of Deep Reinforcement Learning for Cybersecurity", supervised by Dr. Fabio Pierazzi.
- Received funding from a NMES Faculty Studentship.

Visiting Researcher @ S2Lab

UNIVERSITY COLLEGE LONDON

London, UK
January 2024 - Present

- Continuing PhD research with my primary supervisor Dr. Fabio Pierazzi.
- Additionally, aided in wider projects such as the extension of the Tesseract framework and ACSAC Cybersecurity Artifacts Competition.

Visiting Collaborator @ AICD

THE ALAN TURING INSTITUTE

London, UK
October 2023 - Present

- On going collaborations with principal research scientists Dr. Chris Hicks and Dr. Vasilios Mavroudis.
- A member of the DARPA AI Cyber Challenge (AixCC) team, which developed a LLM pipeline for automated vulnerability detection and repair.
- Developed a deep reinforcement learning (DRL)-based autonomous security testing tool for denial-of-service in GraphQL applications.

Graduate Teaching Assistant

KING'S COLLEGE LONDON

London, UK
January 2024 - May 2024

- MSc Security Testing module labs.
- BSc/MSc Network Security module labs.

King's Undergraduate Research Fellow

KING'S COLLEGE LONDON

London, UK
July 2022 - September 2022

- Performed evaluations on the impact of poisoning on machine learning classifiers for malware detection under the effects of concept drift.

Education

B.Sc. (Hons) in Computer Science (Artificial Intelligence)

KING'S COLLEGE LONDON

London, United Kingdom
2020 - 2023

- First Class Honours with a specialization in Artificial Intelligence.
- Dissertation on "Adversarial Machine Learning Evaluation of the MaMaDroid Feature Space", which evaluated the impact of data poisoning alongside different mitigation strategies on a malware classifier over time.

Papers

PUBLICATIONS

2026	SoK: The Pitfalls of Deep Reinforcement Learning for Cybersecurity, Shae McFadden , Myles Foley, Elizabeth Bates, Ilias Tsingenopoulos, Sanyam Vyas, Vasilios Mavroudis, Chris Hicks, Fabio Pierazzi. [paper], [code].	USENIX Security
2026	DRMD: Deep Reinforcement Learning for Malware Detection under Concept Drift, Shae McFadden , Myles Foley, Mario D'Onghia, Chris Hicks, Vasilios Mavroudis, Nicola Paoletti, Fabio Pierazzi. [paper], [code].	AAAI
2026	One Pic is All it Takes: Poisoning Visual Document Retrieval Augmented Generation with a Single Image, Ezzeldin Shereen, Dan Ristea, Shae McFadden , Burak Hasircioglu, Vasilios Mavroudis, Chris Hicks. [paper], [code].	TMLR
2026	Direction for Detection: A Survey of Automated Vulnerability Detection and all of its Pain Points, Dan Ristea, Shae McFadden , Ezzeldin Shereen, Madeleine Dwyer, Sanyam Vyas, Chris Hicks, Vasilios Mavroudis. [paper].	TMLR
2024	The Impact of Active Learning on Availability Data Poisoning for Android Malware Classifiers, Shae McFadden , Zeliang Kan, Lorenzo Cavallaro, Fabio Pierazzi. [paper], [code].	ARTMAN co-located with ACSAC

2024	Wendigo: Deep Reinforcement Learning for Denial-of-Service Query Discovery in GraphQL , Shae McFadden , Marcello Maugeri, Chris Hicks, Vasilios Mavroudis, Fabio Pierazzi. [paper], [code].	<i>DLSP co-located with IEEE S&P</i>
2023	Poster: RPAL-Recovering Malware Classifiers from Data Poisoning using Active Learning , Shae McFadden , Zeliang Kan, Lorenzo Cavallaro, Fabio Pierazzi. [paper], [code].	<i>ACM CCS Poster</i>

NON-ARCHIVAL

2026	Building Better Environments for Autonomous Cyber Defence , Chris Hicks, Elizabeth Bates, Shae McFadden , Isaac Symes Thompson, Myles Foley, Ed Chapman, Nickolas Espinosa Dice, Ankita Samaddar, Joshua Sylvester, Himanshu Neema, Nicholas Butts, Nate Foster, Ahmad Ridley, Paul Jones.	<i>AI4CNI co-located with AAMAS</i>
2025	One Pic is All it Takes: Poisoning Visual Document Retrieval Augmented Generation with a Single Image (Short Paper) , Ezzeldin Shereen, Dan Ristea, Burak Hasircioglu, Shae McFadden , Vasilios Mavroudis, Chris Hicks.	<i>MAGMaR co-located with ACL</i>
2024	Impact of "TESSERACT: Eliminating Experimental Bias in Malware Classification across Space and Time" , Shae McFadden , Zeliang Kan, Daniel Arp, Feargus Pendlebury, Roberto Jordaney, Johannes Kinder, Fabio Pierazzi, Lorenzo Cavallaro. [paper], [code].	<i>ACSAC Artifacts Competition</i>

ARXIV

2026	REPLICANT: Learning Policies for Evading and Hardening Malware Detectors , Shae McFadden , Ilias Tsingenopoulos, Mario D’Onghia, Alexander Herzog, Myles Foley, Chris Hicks, Lorenzo Cavallaro, Fabio Pierazzi. [paper].	<i>arXiv</i>
2024	TESSERACT: Eliminating Experimental Bias in Malware Classification across Space and Time (Extended Version) , Zeliang Kan, Shae McFadden , Daniel Arp, Feargus Pendlebury, Roberto Jordaney, Johannes Kinder, Fabio Pierazzi, Lorenzo Cavallaro. [paper], [code].	<i>arXiv</i>

Talks & Presentations

The 35th USENIX Security Symposium		<i>Baltimore, USA</i>
PRESENTER FOR <SOK: THE PITFALLS OF DEEP REINFORCEMENT LEARNING FOR CYBERSECURITY>.		2026
Presented a long presentation and poster for our paper published at USENIX Sec on the common issues, or pitfalls, in published deep reinforcement learning research for cybersecurity.		
The Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)		<i>Singapore</i>
PRESENTER FOR <DRMD: DEEP REINFORCEMENT LEARNING FOR MALWARE DETECTION UNDER CONCEPT DRIFT>.		2026
Presented the poster and video (link here) for our paper published at AAAI on the use of deep reinforcement learning to improve the over time performance of malware detection under concept drift.		
Recent Advances in Resilient and Trustworthy Machine Learning Workshop (ARTMAN) co-located with ACSAC		<i>Hawaii, USA</i>
PRESENTER FOR <THE IMPACT OF ACTIVE LEARNING ON AVAILABILITY DATA POISONING FOR ANDROID MALWARE CLASSIFIERS>.		2024
Presented my paper published at the workshop on the use of active learning to recover malware classifiers from poisoning.		
Deep Learning Security & Privacy Workshop (DLSP) co-located with IEEE S&P		<i>San Francisco, USA</i>
PRESENTER FOR <WENDIGO: DEEP REINFORCEMENT LEARNING FOR DENIAL-OF-SERVICE QUERY DISCOVERY IN GRAPHQL>.		2024
Presented my paper published at the workshop on the use of DRL to discover low-rate DoS queries in GraphQL applications.		
Machine Learning and Cyber Security Symposium @ Imperial		<i>London, UK</i>
PRESENTER FOR <DEEP REINFORCEMENT LEARNING FOR DENIAL-OF-SERVICE QUERY DISCOVERY IN GRAPHQL>.		2024
Presented my work on using deep reinforcement learning to discover low-rate denial-of-service queries in GraphQL applications.		
WorldCUR*BCUR		<i>Warwick, UK</i>
PRESENTER FOR <ADVERSARIAL MACHINE LEARNING EVALUATION OF THE MAMADROID FEATURE SPACE>.		2023
Presented my work on the robustness of the MaMaDroid feature space and its interplay with concept drift mitigation techniques.		

Honours & Awards

COMPETITIONS

2026	European Defence Tech Hackathon , Vigil-Fusion Team. Zero-shot drone detection via multimodal fusion of open-vocabulary vision and audio models.	<i>London, UK</i>
2025	European Defence Tech Hackathon , MARSim Team. MARL environment for simulating both UAV attack and defence swarms during UGV resupply missions.	<i>London, UK</i>

2025	LLMail-Inject: Adaptive Prompt Injection Challenge , Mindrake Team. <i>IEEE SaTML</i> .	<i>Copenhagen, DEN</i>
2024	Cybersecurity Artifacts Competition ACSAC , Finalist, ACSAC.	<i>Hawaii, USA</i>
2024	AlxCC Semifinal Competition , Mindrake Team, <i>DEFCON 32</i>	<i>Las Vegas, USA</i>

AWARDS

2025	Top Reviewer , awarded to recognize a reviewer’s professionalism and diligence, <i>ACM AISec</i> .	<i>Taipei, Taiwan</i>
2024	Top Reviewer , awarded to recognize a reviewer’s professionalism and diligence, <i>ACM AISec</i> .	<i>Salt Lake City, USA</i>
2023	Layton Research Award , given not merely for success in passing examinations but for the best promise of aptitude and genius for original scientific work, <i>King’s College London</i> .	<i>London, UK</i>
2023	Alan Fairbourn Memorial Prize , awarded to the student who produces the most meritorious final year project in the Department of Informatics, <i>King’s College London</i> .	<i>London, UK</i>
2023	Associateship of King’s College , elected by the Academic Board of King’s College London as an ‘Associate of King’s College’ (AKC), <i>King’s College London</i> .	<i>London, UK</i>
2020	Governor General Academic Medal , awarded to the student graduating with the highest grade point average from a Canadian high school, <i>University of Winnipeg Collegiate</i> .	<i>Winnipeg, CAN</i>

Academic Service

2026	33rd ACM Conference on Computing and Communication Security (CCS) , PC Member	<i>Netherlands</i>
2026	40th AAAI Conference on Artificial Intelligence , PC Member	<i>Singapore</i>
2025	4th Machine Learning and Cyber Security Symposium (ML-CSS) , Organizer	<i>London, UK</i>
2025	18th ACM Workshop on Artificial Intelligence and Security (AISec) , PC Member	<i>Taipei, Taiwan</i>
2025	4th Workshop on Rethinking Malware Analysis (WoRMA) , PC Member	<i>Venice, Italy</i>
2025	8th Deep Learning Security and Privacy Workshop (DLSP) , PC Member	<i>San Francisco, USA</i>
2024	17th ACM Workshop on Artificial Intelligence and Security (AISec) , PC Member	<i>Salt Lake City, USA</i>
2024	21st Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA) , Sub-reviewer	<i>Lausanne, Switzerland</i>
2024	33rd USENIX Security Symposium , Sub-reviewer	<i>Philadelphia, USA</i>

Technical Skills

Python Programming

PROJECTS USING PYTHON:

- Published several papers which used Python to perform machine learning evaluations.
- Current maintainer of four open-source Python Github repositories (DRMD [link], Wendigo [link], RPAL [link], and Tesseract [link]).

Machine Learning & Deep Learning

PROJECTS USING MACHINE LEARNING:

- Several publications applying machine learning (either supervised or deep reinforcement learning) to cybersecurity problems.
- Undergraduate Dissertation focusing on the safe application of supervised learning to malware classification.

Tech Stack

EXPERIENCE FROM PRIOR PROJECTS

- Python, PyTorch, Scikit-Learn, CleanRL, Stable-Baselines3, Gymnasium, Hugging Face Transformers, NumPy, Git, Docker

Leadership & Teamwork

EXPERIENCES THAT HAVE DEVELOPED LEADERSHIP AND TEAMWORK SKILLS:

- Collaborating with researchers from various institutions on research projects.
- Leading 4 to 10 person teams on software development projects in undergraduate studies.
- Captain of a 12 person fencing team.

Extracurricular Activity

Men’s First Fencing Team (King’s College London)

London, UK

CAPTAIN

2022 - 2024

- Led the team to winning the premier league and the national championships in 2023.

Canadian National Fencing Team

Canada

MEMBER

2017 - 2020

- World Fencing Grand Prix in Montreal, Canada, 2020. U18 World Championships in Torun, Poland, 2019. U18 Fencing Pan American Championships in Bogata, Columbia, 2019. Commonwealth Fencing Championships in Canberra, Australia, 2018.